

Data Types, Data Display and Summary Statistics

TYPES OF DATA

Types of Data

- Quantitative or Qualitative?
 - Quantitative: presented as numbers permitting arithmetic
 - Interest rate
 - Temperature
 - Qualitative (categorical): everything else
 - Country of birth
 - Supplier

Types of Data

- Univariate or Multivariate?
 - Univariate: one fact for each object in a dataset (“one column in a spreadsheet”)
 - Multivariate: two or more facts for each object in a dataset (“many columns in a spreadsheet”)

Types of Data

- Discrete or Continuous?
 - Discrete: counted
 - Cars sold
 - Number of children
 - Continuous: measured (always allow “in-between” values)
 - Gallons of oil sold
 - Temperature
- What about age? Money?

Types of Data

- Ordinal Data

- Definition: “Qualitative data that has an ordering”
- Example – Likert Scale:

disagree strongly < disagree < neutral < agree < agree strongly

- Often “measure” with numbers:

1 = disagree strongly

2 = disagree

⋮

5 = agree strongly

Types of Data

- Nominal Data
 - Definition: “Qualitative data that has no ordering”
 - Example – Car Manufacturer:

Nissan, Honda, Chevy, Ford, Mazda, Toyota, ...

Types of Data

- Time Series or Cross-Sectional?
 - Time series: when time sequencing is important
 - US historical inflation rates
 - A baby's weight
 - Cross-sectional: data are contemporaneous, or time doesn't matter
 - 2025 inflation rates for several countries
 - Weight at birth

Population vs Sample

- Population
 - Entire Set of Observations
 - The entire population is assumed to be fixed.
 - Often hypothetical since we generally cannot collect the entire population
 - However, must be defined
- Sample
 - A subset of observations from the population
 - Easier to collect
 - Sample needs to be representative of the population.
 - Ultimately, we want to infer something about the population.

Random Sample

- Random Sample:
 - All elements of the population have the same chance of being chosen in the sample.
- Hence, our sample will be representative of the population.
- There are various types of random sampling procedures.

Population vs Sample

- Population is Fixed
- Sample is Random
- Parameter: Numerical Summary Measure of Population
- Statistic: Numerical Summary Measure of Sample
- Parameters are fixed while Statistics are ***RANDOM.***

DESCRIPTIVE STATISTICS: GRAPHICAL SUMMARIES

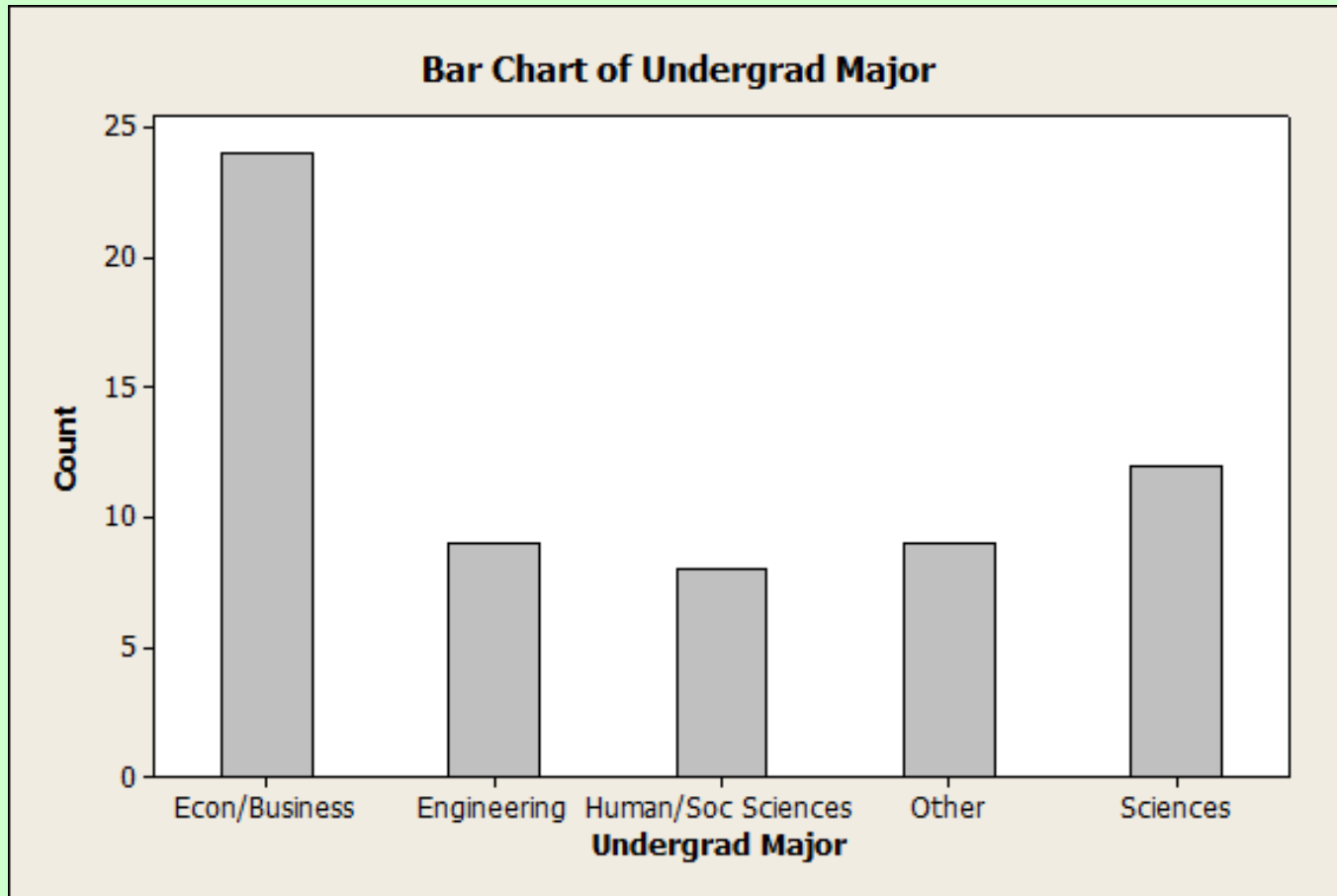
Data Display

- In an MBA class, we know about each student:
 - First name
 - Last name
 - Age
 - Undergraduate major
 - Work experience (in years)
 - Annual Compensation

Dataset

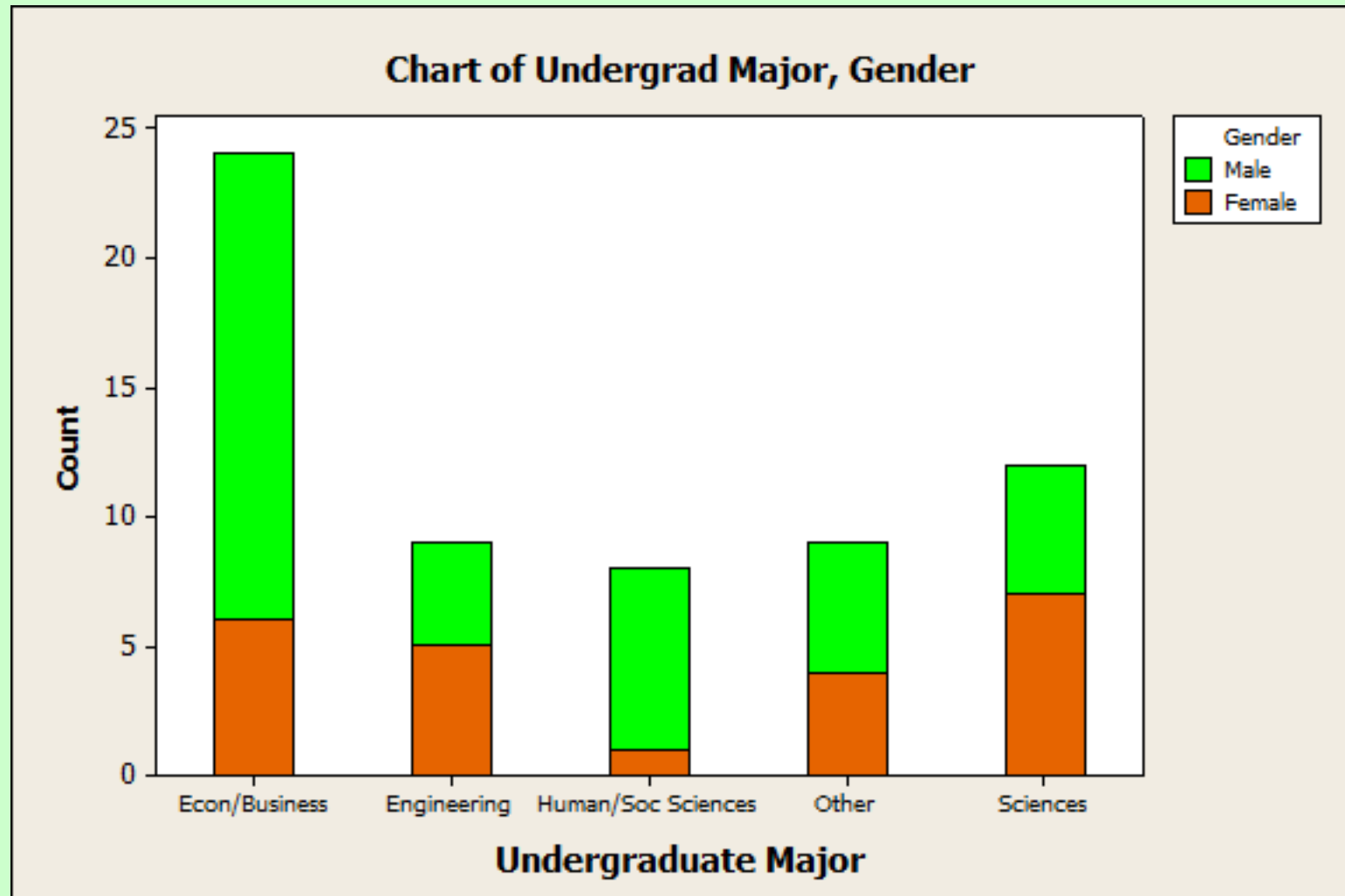
Undergrad Major	Compensation	Age	Years Experience	Gender	Undergrad Major	Compensation	Age	Years Experience	Gender
Econ/Business	106.27	26.72	3.74	Male	Human/Soc Sciences	76.61	23.83	3.37	Male
Econ/Business	96.45	25.88	3.29	Female	Human/Soc Sciences	74.38	30.11	9.89	Male
Econ/Business	103.80	24.16	3.04	Male	Human/Soc Sciences	94.28	30.22	7.85	Male
Econ/Business	107.67	27.27	6.65	Female	Human/Soc Sciences	86.28	30.20	8.36	Male
Econ/Business	101.75	31.13	12.53	Male	Human/Soc Sciences	90.59	31.34	12.99	Male
Econ/Business	135.88	35.81	19.60	Male	Human/Soc Sciences	78.86	29.74	4.10	Female
Econ/Business	109.43	24.68	4.55	Male	Sciences	101.47	29.04	7.88	Female
Econ/Business	100.18	28.94	4.03	Male	Sciences	113.58	32.53	9.89	Male
Econ/Business	112.18	29.74	8.65	Female	Sciences	106.16	31.23	10.73	Male
Econ/Business	113.95	32.07	7.59	Female	Sciences	94.16	30.58	5.49	Male
Econ/Business	118.20	37.87	19.44	Male	Sciences	90.53	27.16	4.87	Female
Econ/Business	94.48	26.91	4.49	Male	Sciences	115.19	27.67	4.93	Female
Econ/Business	109.64	25.72	3.54	Male	Sciences	100.29	25.42	2.16	Female
Econ/Business	117.52	28.41	4.47	Male	Sciences	113.11	26.25	5.95	Male
Econ/Business	115.51	27.53	6.69	Female	Sciences	108.50	31.85	10.19	Female
Econ/Business	112.76	26.36	7.11	Male	Sciences	124.41	33.19	13.27	Female
Econ/Business	121.77	32.15	10.97	Female	Sciences	121.22	38.95	17.67	Female
Econ/Business	101.03	25.42	4.43	Male	Sciences	117.56	32.43	11.39	Male
Econ/Business	112.34	24.49	4.97	Male	Econ/Business	130.55	26.57	7.99	Male
Econ/Business	108.75	25.40	2.29	Male	Econ/Business	104.04	25.04	3.56	Male
Engineering	89.53	26.37	2.32	Female	Econ/Business	108.65	27.11	4.50	Male
Engineering	93.56	26.39	3.16	Male	Econ/Business	111.94	25.48	3.04	Male
Engineering	102.81	31.24	10.09	Male	Other	86.24	28.90	5.11	Male
Engineering	113.40	37.70	16.73	Female	Other	94.62	29.15	7.85	Female
Engineering	94.17	33.38	10.82	Male	Other	95.64	26.02	4.82	Male
Engineering	92.82	35.51	13.85	Female	Other	95.60	24.40	3.27	Female
Engineering	97.51	34.18	10.07	Female	Other	95.42	27.11	7.30	Female
Engineering	116.65	32.19	17.47	Male	Other	88.77	26.36	5.27	Female
Engineering	97.93	28.11	5.30	Female	Other	88.95	31.50	11.28	Male
Human/Soc Sciences	88.95	26.58	4.67	Male	Other	99.80	35.31	10.89	Male
Human/Soc Sciences	67.67	27.85	4.81	Male	Other	97.18	33.49	10.92	Male

The Bar Chart

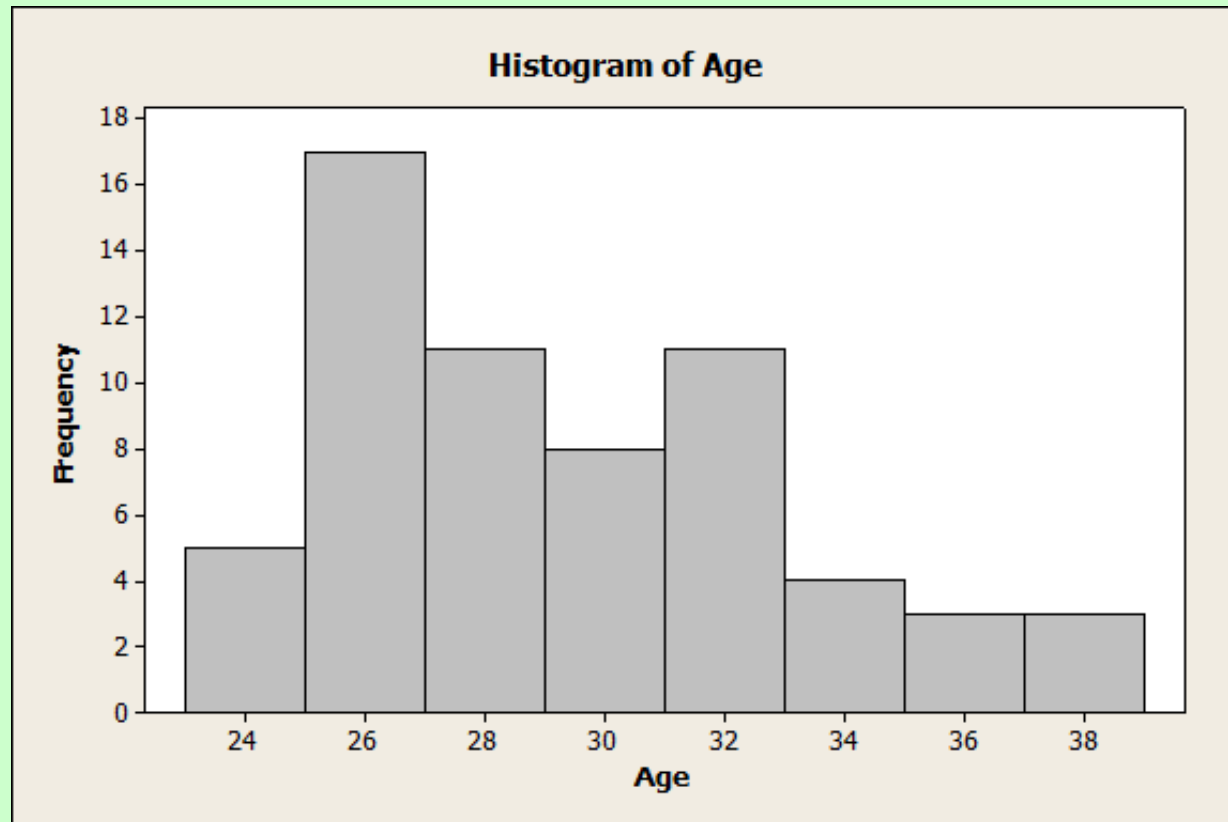


Horizontal axis is qualitative

The Stacked Bar Chart



The Histogram

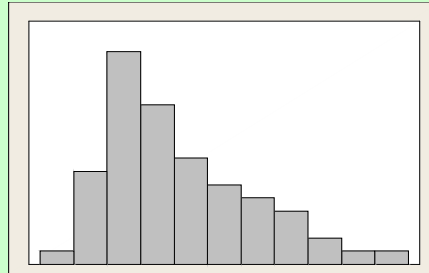


- Horizontal axis is quantitative
- No space between columns

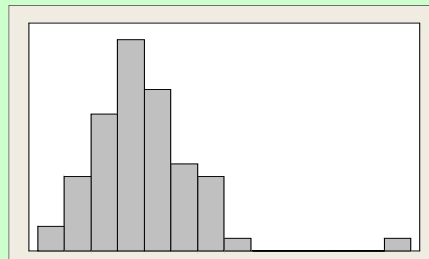
The Histogram

- With a histogram, can detect

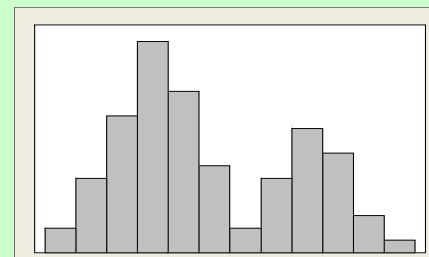
- Skewness:



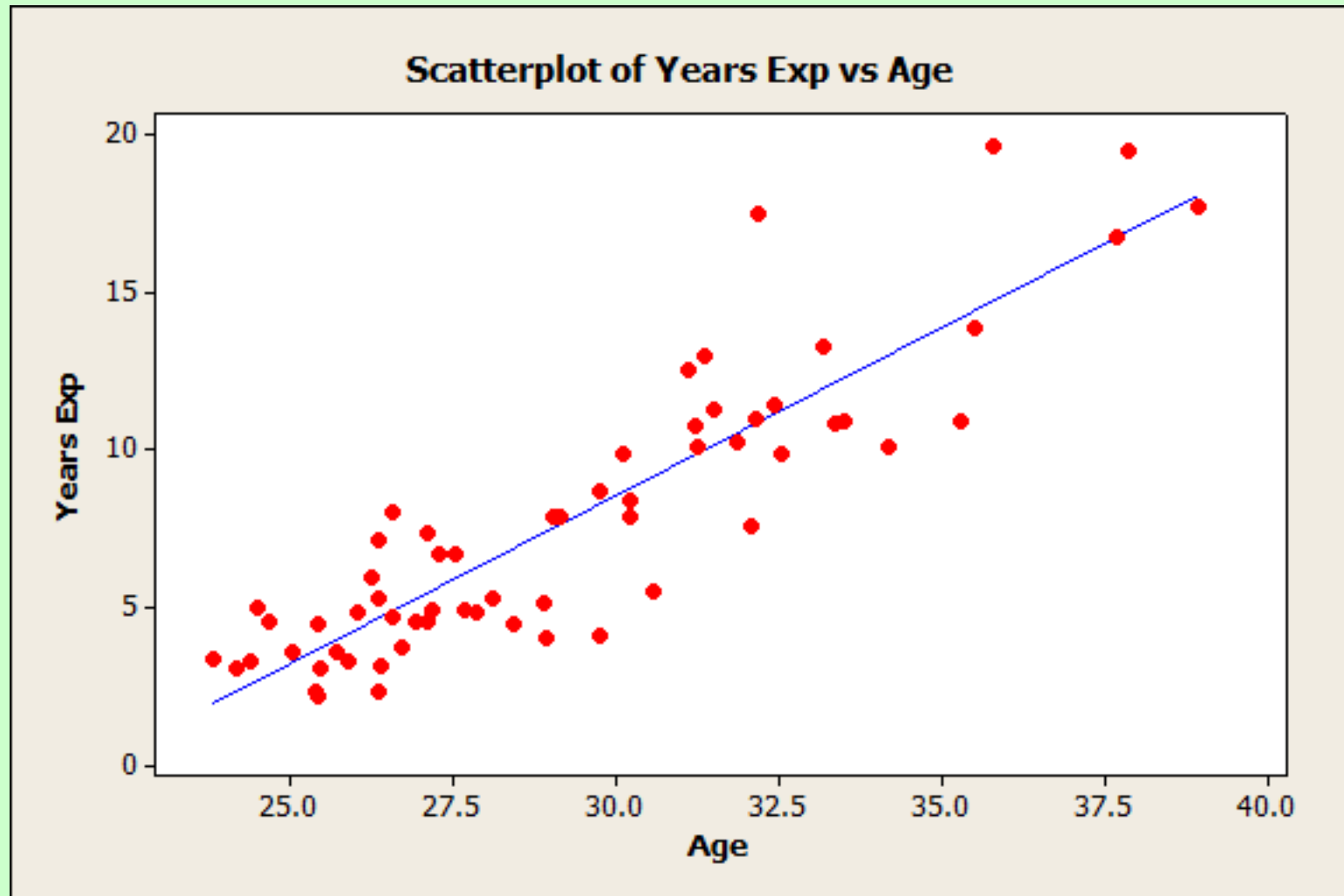
- Outliers:



- Bimodal distribution:

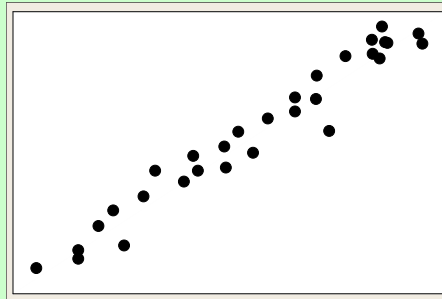


The Scatter Plot

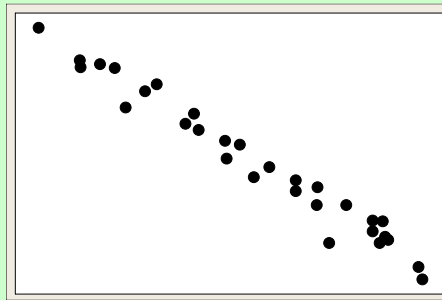


The Scatter Plot

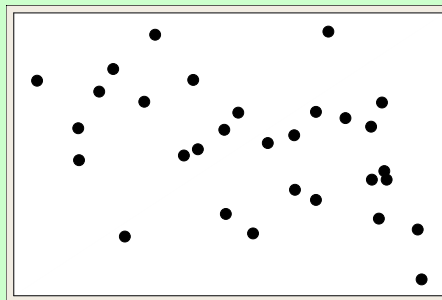
- Positive (linear) relation:



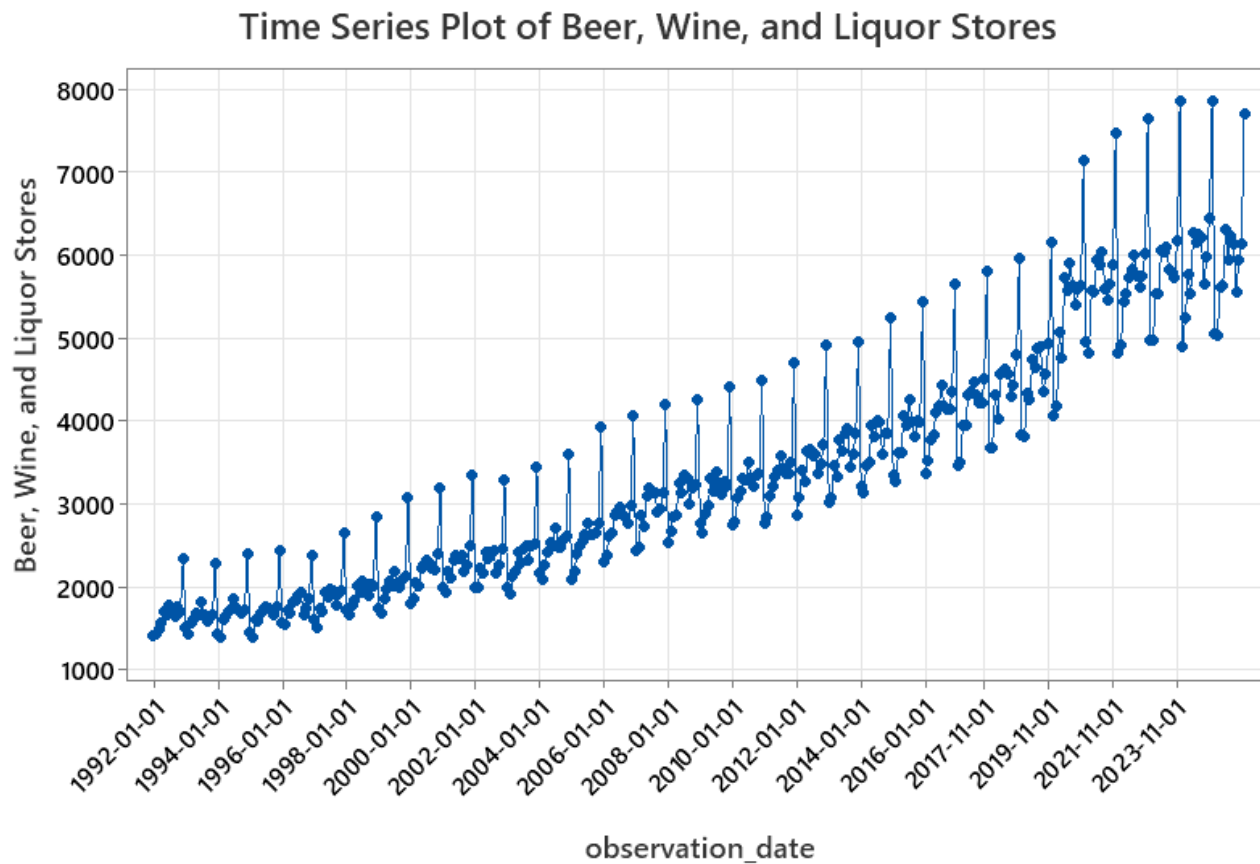
- Negative (linear) relation:



- No relation:



Time Series



DESCRIPTIVE STATISTICS: NUMERICAL SUMMARIES

The Sample Mean

- Compensation Data from MBA students:
Category - Other

- $n = 9$ numbers:

86.24, 94.62, 95.64, 95.60, 95.42, 88.77, 88.95, 99.80, 97.18

x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_n

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_i x_i = \frac{1}{n} \sum x$$

- Sample Mean: 93.58

Population Mean

- $\mu = \frac{\sum_{i=1}^N x_i}{N}$
- Parameter vs Statistic
 - Remember Parameters are Fixed whereas Statistics are Random
- Are there any real ramifications of this difference?

The Median

- Median = the middle value in a *sorted* dataset

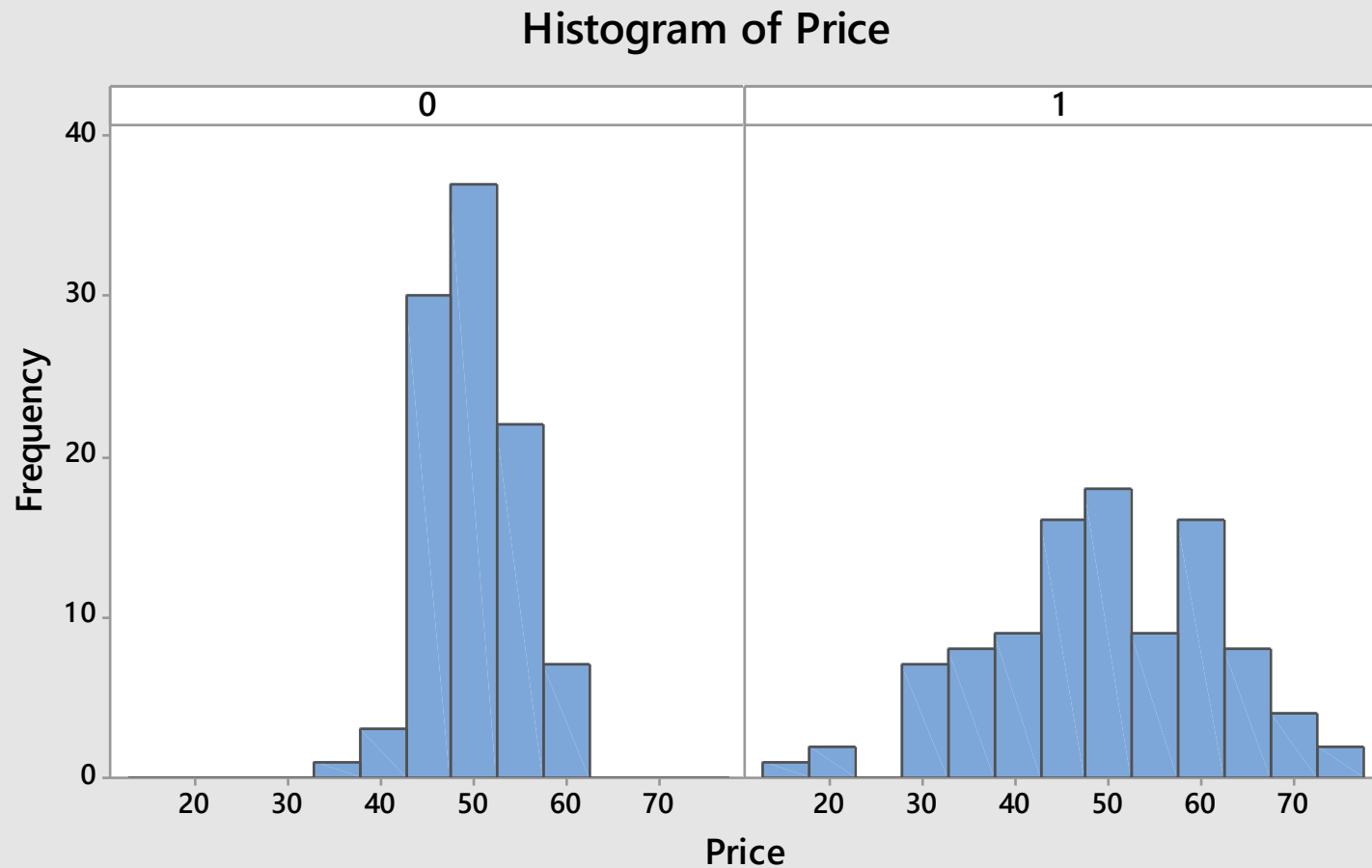
86.24, 88.77, 88.95, 94.62, **95.42**, 95.60, 95.64, 97.18, 99.80

- When n is odd, take “true” middle value
- When n is even, take average of two middle values

Mean vs. Median

- Mean is
 - Sensitive to outliers
 - Includes “more information”
- Median is
 - Useful when ranking is important
 - Important in demographics
- Other “typical value” measures
 - Mode = the most common value
 - Trimmed mean (ignore upper and lower x% of data)

Variability: Is it Good or Bad



Measuring Variability

- Population
- $$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N} = \frac{\sum_{i=1}^N x_i^2 - N\mu^2}{N}$$
- It is a Parameter
- Deviation (from the mean): $x_i - \mu$
- Population Variance is equal to the average squared deviation.
- Good Summary Measure of Variability, But ...

Population Variance and Standard Deviation

- Variance
 - Unit of Measurement is an issue.
 - Consider x as price measured in dollars
 - Then Variance is measured in Dollars Squared.
 - Therefore, we take the square root.
- Population Standard Deviation: $\sigma = \sqrt{\sigma^2}$
- Variance and standard deviation are *always* positive

Sample Variance: Calculating s^2

- Find $\bar{x}$
- For each x_i , calculate $(x_i - \bar{x})$
- In a formula: $s^2 = \frac{1}{n-1} \left(\sum_i (x_i - \bar{x})^2 \right)$
- An easier-to-use formula:

$$s^2 = \frac{1}{n-1} \left(\sum_i x_i^2 - n\bar{x}^2 \right)$$

- To find s , $s = \sqrt{s^2}$

Sample Variance vs Population Variance

- For sample variance, divide by $n - 1$ (rather than by n) when calculating s^2

$$s^2 = \frac{1}{n-1} \left(\sum_i (x_i - \bar{x})^2 \right)$$

- This is because s^2 is random and we want it to be an unbiased estimator of σ^2
- In other words, the average of many different sample variances (each from a different random sample) should be equal to the population variance

Using Mean and Standard Deviation

- Empirical Rule (assuming distribution is somewhat bell shaped and symmetric)
- Approximately 68% of observations are within 1 standard deviation of the mean
- Approximately 95% of observations are within 2 standard deviation of the mean
- Approximately 99% of observations are within 3 standard deviation of the mean

Z-transformation

- Population: $z = \frac{x - \mu}{\sigma}$
- Sample: $z = \frac{x - \bar{x}}{s}$
- Can be used with Empirical Rule (many cases on one graph)
- Outlier Detection:
 - $2 \leq |z| < 3$: Possible Outlier
 - $|z| \geq 3$: Outlier

Quartiles

- First quartile = the value in position $(n+1)/4$ in a sorted dataset
 - Also called Q_L , 25th percentile, Q_1
- Median is the second quartile
- Third quartile = the value in position $3(n+1)/4$ in a sorted dataset
 - Also called Q_U , 75th percentile, Q_3

Measures of Variability

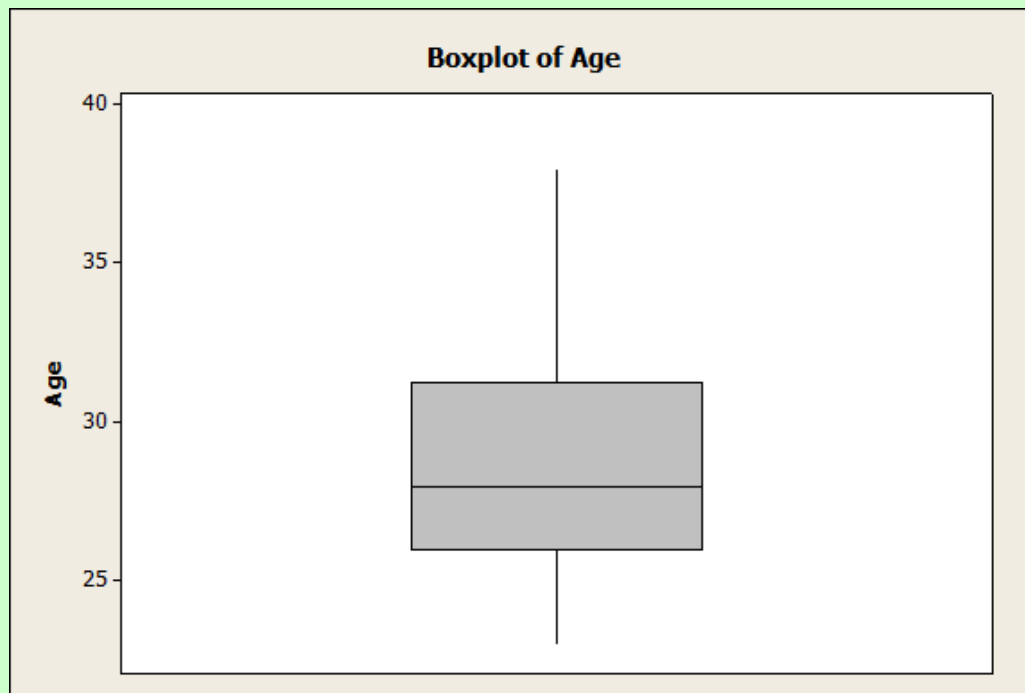
- Range = max – min (very sensitive to outliers)
- $\sigma \equiv$ population standard deviation and $s \equiv$ sample standard deviation (sensitive to outliers, why?)
- IQR = $Q_U - Q_L$ (not sensitive to outliers)
 - Can be used to detect outliers.

The Box Plot

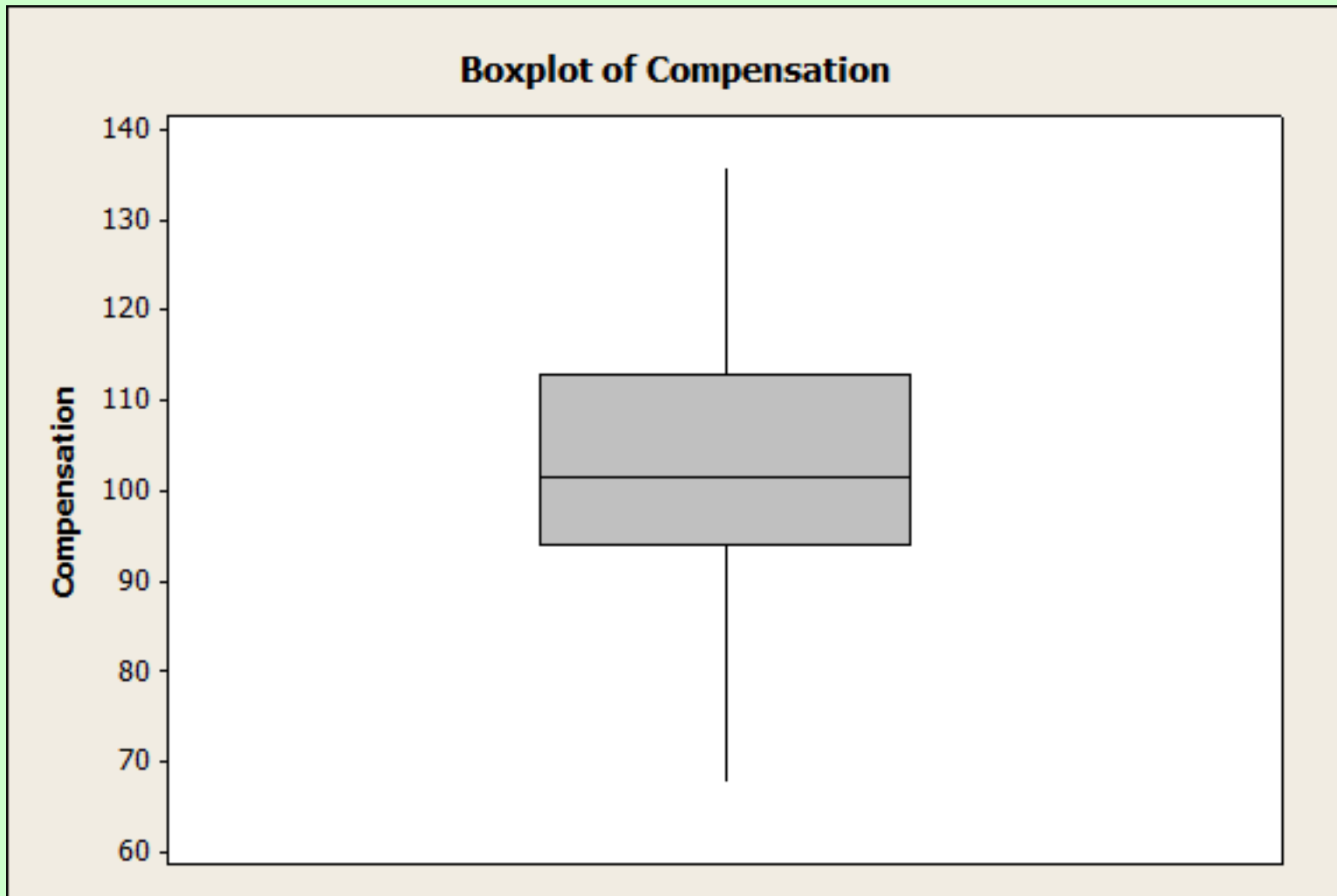
- Inter-Quartile Range (IQR) = $Q_U - Q_L$
- Inner fences:
 - Lower inner fence = $Q_L - 1.5 \text{ IQR}$
 - Upper inner fence = $Q_U + 1.5 \text{ IQR}$
- Outer fences:
 - Lower outer fence = $Q_L - 3.0 \text{ IQR}$
 - Upper outer fence = $Q_U + 3.0 \text{ IQR}$
- Data outside inner fence = outlier
- Data outside outer fence = serious outlier

The Box Plot

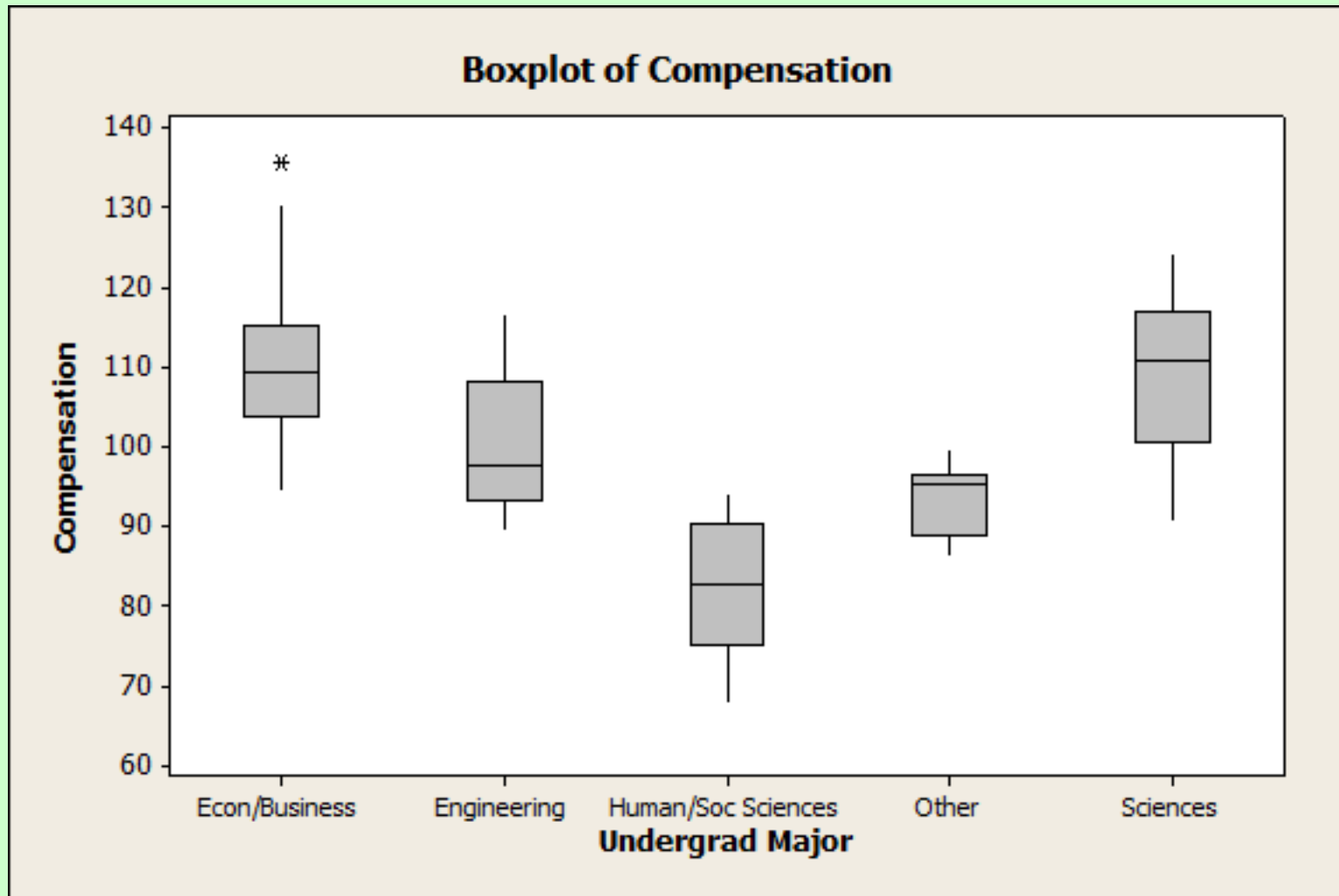
- Box edges (“hinges”) at Q_U and Q_L ; mark median with bar
- Draw “whiskers” from edges to smallest and largest non-outliers



The Box Plot

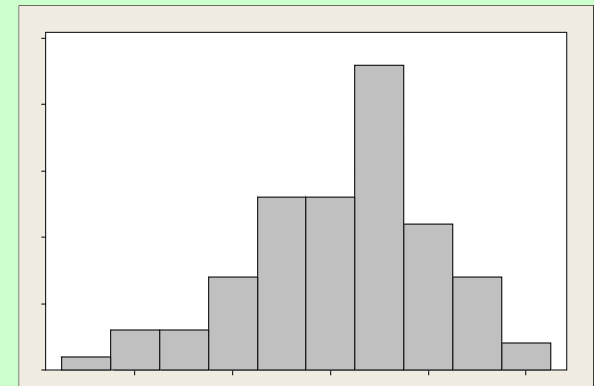
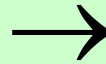
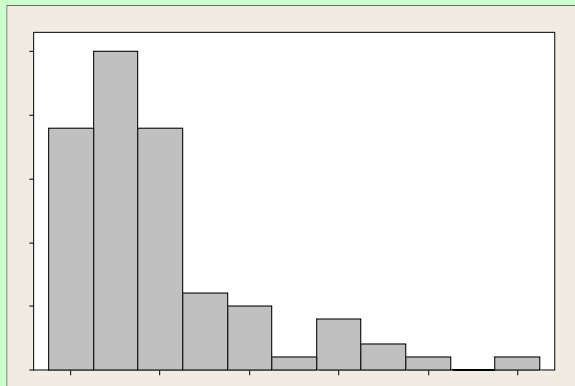


The Box Plot



Transformations

- Often we want to convert a skewed dataset to a more symmetric one



- Can use the *Log transformation*